PERSONAL INFORMATION

Réussite BUGALE MALEMBAKA

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Sex Male | Date of birth 14/06/1993 | Nationality Congolese (DR Congo)

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.EDUCATION

Jan – Feb 2025 — Visiting student, University of Oxford, United Kingdom
Food Systems Group, Environmental Change Institute, University of Oxford, UK.
Exchange on Scientific methods in food systems thinking and footprints

Jan 2022 – Jan 2026 — PhD studies, Environmental systems science (ongoing)
Agroecological transitions group, Department of environmental systems science, ETH Zurich, Switzerland

Sept 2017 – Sept 2020 — Master studies (M.Sc. degree in Soil science)
Department of Land Resource Management and Agricultural Technology, Faculty of Agriculture, University of Nairobi, Kenya

Sept 2011 – July 2016 — Bachelor studies (B.Sc. degree in Agronomy)
Faculty of Agronomic sciences, Catholic University of Bukavu, DR Congo

Sept 2005 - July 2011 — Secondary school at Institut Bwindi, Bukavu, DR Congo
Option Chemistry Biology, State Diplôma

LANGUAGE SKILLS

Language(s)

	LISTENING	READING	SPEAKING	WRITING
English	C1	B2	C1	B2
French	C2	C2	C2	C1
Brazilian Portuguese	B1	B2	B1	A2
German	A2	B1	A2	A1
Swahili	Mother tongue			

Levels: A1/2: Basic user - B1/2: Independent user - C1/2 Proficient user

PUBLICATIONS (Selected)

- **Malembaka, R.B.**, Pfister, S., Thom, B., Cintrão, R.P., Trivilin, M.I. and Jacobi, J., 2025. Are alternatively organized value chains more environmentally sustainable? Evidence from soybean production in Minas Gerais and Paraná states, Brazil. Frontiers in Sustainable Food Systems, 9, p.1681792. https://doi.org/10.3389/fsufs.2025.1681792
- **Malembaka, R.B.** , Pfister, S., Thom, B., Kintche, K., Jacobi, J., 2025: Ecological sustainability and role of Farmers' Organisations in soybean production systems in DR Congo, **preprint** doi: 10.2139/ssrn.5165929 (Submitted to Elsevier Journal: Environmental Impact Assessment Review (in 2nd revision process).
- **Malembaka, R.B.**, Stephan Pfister, Bianca Torres, Johanna Jacobi. 2025. Environmental sustainability of soybean - chicken productions: local and global value chains analysis from Paraná and São Paulo states, Brazil. Under revision in Environ. Res.: Food Syst. Manuscript number: ERFS-100194.
- Cintrão, R.P., Trivilin, M.I., Jacobi, J., **Malembaka, R.B.** and Sigrist, M., 2024. O poder do soja no Paraná e o uso de agrotóxicos: um desafio à agroecologia. *Cadernos de Agroecologia*, *19*(1).

- **Malembaka. R. B**., Onwonga, R., Jefwa, J., Ayuke, F. and Nabahungu, L. 2021. **Diversity and distribution of Arbuscular Mycorrhizal Fungi, and soils inoculum potential in maize (*Zea mays*) cropping systems of South Kivu, DR Congo** African Journal of Agricultural Research. Vol. 17(4), pp. 604-617, DOI: 10.5897/AJAR2020.15390
- **Malembaka. R. B**., Onwonga, R. Jefwa, J., Ayuke F and Nabahungu, L. **Role of selected AMF species on maize (*Zea mays*) performance, phosphorus uptake and roots colonisation in a Ferralsol and a Nitisol under controlled conditions.** Vol. 18, No. 3, p. 20-32, 2021
- **Malembaka R. B.** 2019. Characterization and screening of native arbuscular mycorrhizal fungi isolates from maize (*Zea mays* L.) agro ecosystems in South Kivu, Democratic Republic of Congo. MSc Thesis dissertation. University of Nairobi. 122p
- Co-director of **BEBUC Newsletter no 3. 2020**. BEBUC, a unique and sustainable scholarship program for the renewal of excellence in higher education institutions of the Democratic Republic of the Congo. 19 pages
- **Malembaka R. B.** 2016. **Identification of local rhizobia strains effectives on soybean (*Glycine max*) under greenhouse conditions. BSc dissertation. Catholic University of Bukavu**. 52p (Link to the abstract: https://www.africmemoire.com/read-identification-des-souches-locales-de-rhizobium-effectives-sur-le-soja-glycine-max-en-conditions-de-serre-1891.html)
- **Malembaka R. B.** 2014. **Effet de la fertilisation organique, minérale et phosphorée sur le rendement du niébé (*Vigna unguiculata* L.) dans les conditions édapho-climatiques de Karhale**. Travail de Fin de Cycle. Inédit. UCB. 41 p.

GRANTS, FUNDINGS, SCHOLARSHIPS

- **D-USYS CO2 compensation fund :** Project titled: 'Integration of multipurpose trees for enhancing production and erosion control in small-scale Soybean-Coffee agroforestry farms in Eastern DR Congo'. Action research grant awarded by the Department of Environmental sciences , ETH Zurich. 2025. 11'000 CHF
- **D-USYS CO2 compensation fund**. Project: 'Promotion and joint implementation of diverse agroforestry systems in coffee and soybean systems in South Kivu, DR Congo', Action research grant awarded by the Department of Environmental sciences , ETH Zurich. 2023. 9'700 CHF.
- **Early career grant, Swiss Leading house for Latin America region**. Project: 'Assessment of Ecological footprints of soybean production systems in Minas gerais and Parana states, Brazil'. Swiss Leading house for Latin America region, University of St. Galen. 2023. 4'500 CHF
- **PhD studies position at ETH Zurich. September 2021**. In the project "Deliberate: Linking the deliberative quality of soy and coffee value chains to ecological foodprints", funded by the SNSF (https://agroecological-transitions.ethz.ch/people.html)
- **Travel grant** to attend the FISTAM symposium (First International Symposium on Tropical Mycology: http://fistam.leb-up.org/) in Parakou, **Benin** in September **2019**. Organized by the University of Parakou and the Goethe University Frankfurt. Funded by the Volkswagen Foundation
- **BEBUC Excellence Scholarship** for **Master studies** in Agricultural sciences at University of Nairobi, April **2017**. Funded by *Else Kröner-Fresenius Foundation* via *f*UNIKIN – Förderverein Uni Kinshasa (e.V.).(https://foerderverein-uni-kinshasa.de/?lang=en)
- **BEBUC Excellence Scholarship** for B.Sc. studies, obtained in 2013, with prolongation in 2014 and 2015), Funded by Else Kröner-Fresenius Foundation via *f*UNIKIN – Förderverein Uni Kinshasa (e.V.)

- BSc dissertation **research supported** by the N2Africa project through funds from Bill and Melinda Gates foundation, at the International Institute of Tropical Agriculture (IITA) at Kalambo, South Kivu, DRC, in 2015-2016

WORK EXPERIENCE

Jan 2022 – Now

Doctoral researcher at ETH Zurich:
Fieldwork in Latin America (Brazil) and Africa (DR Congo) in cropping systems and sustainability. Supervision of bachelor students. Attend conference, writing papers for publications. Academic visit in Oxford University

Feb 2017 – Nov 2021

Researcher and Lecturer Assistant at UCB (Université Catholique de Bukavu)

Lecturer Assistant at the Faculty of agricultural sciences Agronomy of the Catholic University of Bukavu, in charge of Practicals of general and tropical Pedology, Plants nutrition and Fertilisation, Topometry and Cartography, Fertilizers Technology, Soil biology units in the bachelor program. Research on Soil fertility, sustainable intensification (good agronomy , improved varieties, organic fertilization, intercropping) of coffee production in South Kivu (in the project SVC (strengthening Value Chains) in partnership with World Coffee Researh, TetraTech and Techno Serve in South Kivu, DR Congo

February 2021 – Dec 2021

Tutor of BEBUC Scholars of Bukavu
I oversaw accompanying the students and pupils of Bukavu beneficiaries of the BEBUC (Bourse d'Excellence Bringmann aux Universités Congolaises) scholarship in their educational development and maintenance of academic excellence

Jan 2020 – June 2020

Agronomist Research consultant at VLIR-UOS at UCB
Responsible for conducting and supervising agronomic research trials under field conditions in Walungu, in South Kivu. Leading a research team of 4 people and supervision of two bachelor students at the Faculty of Agricultural Sciences of UCB. Research title: Assessment of the influence of phosphorus mineral fertilization and arbuscular mycorhizae fungi biofertilizer application on phosphorus use efficiency and crop yield in maize-bean intercropped in field conditions in Walungu, South Kivu

Aug 2015 – Jan 2016

BSc research training/Internship at the International Institute ot Tropical Agriculture (IITA)

Laboratory of microbiology of IITA Kalambo in DR Congo/ N2Africa. I was trained in rhizobia inoculant production, inoculation and nodules observation.

COMMUNICATION AND MANAGERIAL SKILLS

Good communication skills gained through my international experience as an international doctoral researcher at ETH, Oxford, in Brazil and DR Congo. Also, as a Speaker and Tutor of BEBUC Scholars.

JOB-RELATED SKILLS

Sustainability assessment

- Ecological and agronomic inventory of crops productions
- Life cycle assessment of crops and animals' productions and value chains
- Impacts assessments, sensitivity analysis. Food systems value chains thinking

Crops production and soil science

- Development of research proposals to improve crops production and manage sustainably agricultural resources. Conduct agronomic field trials.
- Laboratory analysis of soil physical and chemical properties; mineral fertilizers recommendations formulation and crops production responses assessment

Project management and research coordination

Supervision of bachelor students :

Odette Tcheupe: Effet de la fertilisation organo minérale et phosphatée sur la croissance et le rendement de l'oignon dans les conditions edapho-climatique de Mbobero au Sud Kivu, RDC. UCB. Bachelor dissertation, 2021.

Oscar Balibike. Impact de la ferilisation minerale phosphatee sur la production de l'arachide dans les conditions de Bagira, à Kabare. UCB. Bachelor dissertation, 2021.

Jimmy Nyakasane: Diversité des champignons mycorrhiziens dans les systèmes de culture à base de maïs (*Zea mays*) et le potentiel mycorhizien des sols au Sud Kivu, RD Congo. Mémoire de fin de cycle universitaire. UCB. 2019

Yves Marhegane : Effet de fertilisation potassique sur le rendement des variétés de tomate

Computer skills

- Very good knowledge of GIS with **ArcGIS Pro** and **QGIS** softwares.
- Very good command skills with Python, R and Github: https://github.com/rbugale
- Very good knowledge of Brightway2 and Activity Browser

Conferences, Seminars and other courses attended (Selected)

- Attended the **East Africa Conference on Agroecology**. Nairobi, 2025.
- Attended the **LCA Food Conference 2024**. Healthy food systems for a healthy planet. Universite de Barcelona, Spain, 8-12 Sept. 2024
- Attended the **Tropentag 2023**. Competing pathways for equitable food systems transformation: trade-offs and synergies. Humboldt Universitat Zu Berlin, Germany
- **Brightcon 2023**: Open sustainability event - Brightway. 16-20/09/2023, Esche-Sur-Azette, Luxembourg.
- Attended the World Food System Summer school 2022. **Food Systems in Transition**, 6 – 21, August 2022, Rheinau, Switzerland
- Attended the **First International Symposium on Tropical Africa Mycology**, Parakou University, 9-13 September, 2019, Parakou, Benin

REFEREES

Prof. Dr Johanna Jacobi
Department of Environmental Systems Science D-USYS, ETH Zurich, Switzerland
Email: Johanna.jacobi@usys.ethz.ch ; Phone +41 44 632 47 53

Prof Dr Johan Six
Department of Environmental Systems Science D-USYS, ETH Zurich, Switzerland
Email: johan.six@usys.ethz.ch ; Phone +41 44 632 84 83

Dr Stefan Pfister
Ecological design group, ETH Zurich
stephan.pfister@ifu.baug.ethz.ch

Prof. Dr Richard Onwonga
Faculty of Agriculture, University of Nairobi, Kenya. Master thesis supervisor
Email : richard.onwonga@uonbi.ac.ke ; Phone : +254 725 828 254

Done in Zurich, 7th November 2025,
Réussite Bugale Malembaka